

# ENCI335

## Structural Analysis & Systems I

### Part II

## Analysis of Indeterminate Structures

### Course Outline

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- Analysis goal: compute the **demands** for a loaded structure
  - Displacement demands: **deflection** ( $u, v$ ) and **rotation** ( $\theta$ )
  - Internal force demands: **bending moment** ( $M$ ), **shear force** ( $V$ ), and **axial force** ( $N$ )
- Develop the important skill of visualising and computing the manner in which structures respond under the application of external loads.
- Provides an overview of different structural systems and discusses techniques to visualise and compute their response to static loads.
- Covers statically determinate and indeterminate analysis techniques, energy methods, the principle of virtual work, and the flexibility method of analysis.

# Learning Objectives

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- Understand the **force and displacement methods** of analysis and apply them to analyse indeterminate structural systems. (*EIE3, EIE4, WA1*)
- Understand the **energy concepts** and use them to analyse statically determinate and indeterminate structural systems. (*EIE3, BICC1, WA1*)
- Understand the concept of **unit load method** and its application in analysing statically indeterminate structures. (*EIE3, EIE4, WA1*)

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- Module 1: Force/Flexibility method for statically indeterminate beams
- Module 2: Determinate and Indeterminate Frames
- Module 3: Energy methods and virtual work principle
- Module 4: Introduction to structural forms of buildings (when time permits)

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- ENGR 102: Engineering Mechanics.
  - The notions of force-balance and equilibrium
  - Analysis of statically determinate trusses using the section and joint methods
- ENCN 231: Solid Mechanics.
  - Plotting shear force and bending moment diagrams for statically determinate beams
  - Calculating deflections of beams using direct integration and moment-area theorems

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- Notes on whiteboards (bring your notebook)
- Notes shared on Learn
- Reference books, e.g. Beer et al. - Mechanics of Materials, 6th edition or newer.

